

Installing Vivado & Programming the Basys 3 FPGA Board

A complete step-by-step setup guide

1. Installing Vivado 2026.1

Windows Security / Firewall Prompt

During installation, Windows may show a prompt: "**Do you want to allow public and private networks to access this app?**" for OpenJDK Platform binary.

Click **Allow**. This is expected—Vivado's installer uses a bundled Java runtime and this permission is required for the installer/tools to function correctly.

Selecting the Product

On the **Select Product to Install** screen, choose **Vivado**. This is the correct choice for FPGA design work: writing/simulating HDL, synthesis, implementation, and generating bitstreams.

(Choose Vitis instead only if you also need embedded software development or High-Level Synthesis on top of Vivado.)

Customizing the Installation

On the **Vivado Design Suite** customization screen:

- Keep **Vivado** and **Vitis HLS** checked (core tools).
- Under **Devices** → **7 Series**, check **Artix-7 FPGAs** — this is the FPGA family used on the Basys 3 board.
- Leave UltraScale, UltraScale+, Versal, SoCs, Kria, and Alveo unchecked to save disk space (not needed for Basys 3).
- Keep **Install Cable Drivers** and **Acquire or Manage a License Key** checked under Installation Options.

Accepting License Agreements

Check every **"I Agree"** checkbox shown (Vivado EULA, DocNav EULA, Model Composer EULA, and any Third-Party Software EULAs). These are standard legal agreements required to proceed.

Destination Directory

The default install location (C:\AMDDesignTools) and default shortcut/file-association options are fine to leave as-is, provided you have enough free disk space (Vivado + Artix-7 support typically requires ~70–90 GB).

WinPcap Driver Prompt

A separate installer window for **WinPcap** may appear during installation. This is a packet-capture driver bundled for certain networking/debug features. Leave the checkbox checked and click **Install** — it is safe and standard.

"Pre/Post Installation Tasks Failed" Warning (Cable Drivers)

You may see a warning that says the cable driver installation failed. This does **not** mean Vivado failed to install — only the USB/JTAG cable driver step needs a manual retry. Click **OK** to dismiss and continue; Vivado itself is fully installed.

Manually Installing Cable Drivers (Digilent/Basys 3)

Navigate to:

```
C:\AMDDesignTools\2026.1\Vivado\data\xicom\cable_drivers\nt64\install_drivers
```

Right-click **install_drivers** (Windows Command file) and choose **Run as administrator**. Click **Yes** on the User Account Control prompt.

This installs drivers for all supported cable types, including the **digilent** folder needed for the Basys 3's onboard USB-JTAG chip. You do not need to run anything from the digilent subfolder separately.

You do NOT need the Basys 3 plugged in to run the driver installer. Plug it in later, when you're ready to program the board — Windows will then recognize it using the drivers you just installed.

To confirm installation succeeded, open the log file:

```
C:\AMDDesignTools\.xinstall\2026.1\cable_driver_install.log
```

Look for messages like "Driver package added successfully" and no FAILED/ERROR lines near the end.

2. Licensing Vivado (Free "No Charge" Tier)

Vivado 2026.1 uses AMD's tiered licensing system. For the Basys 3 (Artix-7, xc7a35t), you need the free **Vivado Basic Tier License, Node Locked License** (equivalent to the older "WebPACK" license).

Step 1: Find Your Host ID (MAC Address)

Open **Command Prompt** and run:

```
getmac
```

This lists your network adapters. Choose the **Physical Address** of the adapter that is *not* marked "Media disconnected" (i.e., has an active Transport Name). Copy that address, e.g.:

```
08-A1-36-C0-9E-93
```

Step 2: Generate the License

- 1 Go to the AMD Product Licensing site (accessible via the Vivado license error dialog, or search "AMD Vivado license").
- 2 Under **Create New Licenses** → **Certificate Based Licenses**, check **Vivado Basic Tier License, Node Locked License** (Type: *No Charge*).

- 3 Click **Select a host...**, then choose **Add a host...**
- 4 Fill in the host form: **Host Name** (any label, e.g. "Thinkpad"), **Operating System** → Windows 64-bit, **Host ID Type** → Ethernet MAC, **Host ID Value** → paste your MAC address from Step 1.
- 5 Click **Add**, then **Next** to review, then complete the request.
- 6 You'll see a **Congratulations** confirmation page, and the license file will be emailed to your registered email address from xilinx.notification@entitlenow.com.

Step 3: Install the License File

- 1 Download the .lic attachment from the email and save it somewhere memorable (e.g. C:\Xilinx\licenses\).
- 2 Open Start Menu → search "environment variables" → open "Edit the system environment variables".
- 3 Click **Environment Variables...**
- 4 Under **User variables**, click **New...** and enter: Variable name = XILINXD_LICENSE_FILE, Variable value = full path to your .lic file.
- 5 Click OK on all windows, then restart your PC.
- 6 Open Vivado again — the license error should be gone.

3. Creating a New Vivado Project

- 1 Open Vivado (Desktop shortcut, Start Menu search "Vivado", or run C:\AMDDesignTools\2026.1\Vivado\bin\vivado.bat).
- 2 Click **Create Project** and click through the wizard (Next on the welcome screen).
- 3 On **Project Type**, keep **RTL Project** selected (the default and correct choice for standard HDL design work).
- 4 On the device/part selection screen, search for the exact part on your Basys 3 board: xc7a35tcpg236-1.
- 5 Click Next, then **Finish** to create the project.

The part number xc7a35tcpg236-1 exactly matches the chip printed on the Basys 3 box (XC7A35T-1CPG236C): Artix-7 family, 35K logic cells, cpg236 package, speed grade 1.

4. Writing a Simple Blinking-LED Design

Add a Verilog Source File

- 1 In the Flow Navigator, click **Add Sources**.
- 2 Choose **"Add or create design sources"** → Next.
- 3 Click **Create File**, choose type **Verilog**, name it e.g. top.
- 4 Keep file location set to **<Local to Project>** (keeps the file inside your project folder).

- 5 On the "Define Module" screen, add two I/O ports: `clk` (direction: input) and `led` (direction: output). Leave Bus unchecked for both.
- 6 Click OK — Vivado generates a `top.v` file with an empty module skeleton.

Write the Blink Logic

Double-click **top (top.v)** under Design Sources to open it in the editor. Replace the blank body with:

```
module top(  
    input clk,  
    output led  
);  
  
    reg [25:0] counter = 0;  
  
    always @(posedge clk) begin  
        counter <= counter + 1;  
    end  
  
    assign led = counter[25];  
  
endmodule
```

This uses a 26-bit counter that increments every clock edge. On the Basys 3's 100 MHz clock, bit 25 toggles about twice a second, producing a visible blink. Save with **Ctrl+S**.

Add the Constraints File

- 1 In the Flow Navigator, click **Add Sources** → **"Add or create constraints"** → Next.
- 2 Click **Create File**, name it e.g. `constraints` (saved as `.xdc`). Keep it local to project, click OK, then Finish.
- 3 Double-click the new `.xdc` file to open it and paste in the pin mapping below.

```
## Clock signal (100 MHz oscillator)  
set_property PACKAGE_PIN W5 [get_ports clk]  
set_property IOSTANDARD LVCMOS33 [get_ports clk]  
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]  
  
## LED (LD0)  
set_property PACKAGE_PIN U16 [get_ports led]  
set_property IOSTANDARD LVCMOS33 [get_ports led]
```

Pin W5 is the Basys 3's onboard 100 MHz clock; pin U16 is LED0. Save with **Ctrl+S**.

5. Building and Programming the Board

Run Synthesis

In the Flow Navigator, under **Synthesis**, click **Run Synthesis**. Accept default job settings. When it completes, a dialog offers next steps — select **Run Implementation** and click OK.

Run Implementation

Implementation runs automatically if selected above, or can be run manually via **Run Implementation** in the Flow Navigator. Check the **Timing** report tab afterward — "0" failing endpoints under Setup, Hold, and Pulse Width confirms your design meets timing.

Generate Bitstream

Go to **Flow** → **Generate Bitstream** (or find it under **Program and Debug** in the Flow Navigator, which may require scrolling down). This produces the .bit file that gets loaded onto the physical FPGA.

Connect the Basys 3 Board

Plug a USB cable into the Basys 3's USB-JTAG port and the other end into your computer.

The Basys 3 has no separate power switch — it is powered directly through the USB cable. Plugging it in turns it on; you should see a power LED light up automatically.

Program the Board

- 1 Go to **Flow** → **Open Hardware Manager** (or find it under Program and Debug in the Flow Navigator).
- 2 Click **Open target** → **Auto Connect**. Vivado should detect the Basys 3.
- 3 Click **Program device**, select your generated .bit file, and click **Program**.
- 4 Once programming completes, LED0 on the board should begin blinking.